Block practice · Trigonometric identities and equations

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Turning sin and cos equations into tan

Trig equations that factorise

Quadratic trig equations using an identity

OLevels.com SCHOOL

Practice workbook · Turning sin and cos equations into tan

Rearranging an equation with sine and cosine of the same angle (possibly after expanding compound angles) into tan = k, then solving.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-25

1. Question 2 · part whole · 5 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part whole only.

2 Solve the equation

$$\sin(2\theta - 20)° - \sqrt{3}\cos(2\theta - 20)° = 0 \quad \text{for } 0 \leqslant \theta \leqslant 180$$

(5)

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1. Question 2 · part whole · 5 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 5 marks)

1. Question 2 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part b · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

10 (a) Solve the equation

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} \qquad \text{for } 0 \leqslant x \leqslant 2\pi$$

Give your solutions in terms of π, where appropriate.

(3)

(b) Solve the equation

$$3\sin\theta + 5\cos\theta = 0 \qquad \text{for } -360° \leqslant \theta \leqslant 360°$$

Give your solutions to the nearest degree.

(3)

(c) Solve the equation

$$1 + \sin 2y = 2\cos^2 2y \qquad \text{for } -180° \leqslant y \leqslant 0°$$

(5)

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2. Question 10 · part b · 3 marks · Working and answer

Working and answer

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3. Question 10 · part a · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

10

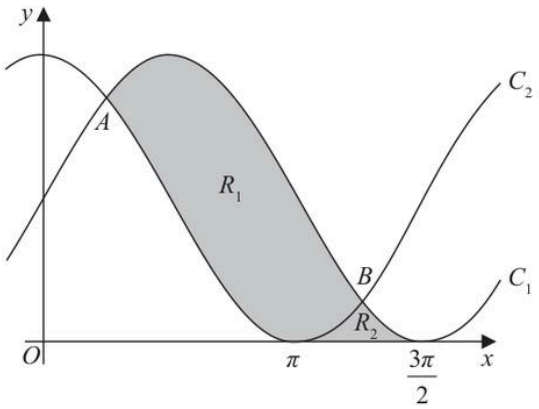


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve C_1 with equation $y = \sin x + 1$ and part of the curve C_2 with equation $y = \cos x + 1$

As shown in Figure 2, C_1 and C_2 intersect at the point A and at the point B

(a) Find the exact value of the x coordinate of A and the exact value of the x coordinate of B **(3)**

The shaded finite region R_1 shown in Figure 2 is bounded by C_1 and C_2

The shaded finite region R_2 shown in Figure 2 is bounded by the x-axis, C_1 and C_2

(b) Use calculus to find the ratio

$$\text{area of } R_1 : \text{area of } R_2$$

Give your answer in the form $a : \left(\frac{\pi\sqrt{2}}{b} - c\right)$ where a, b and c are integers.

(9)

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3. Question 10 · part a · 3 marks · Working and answer

Working and answer

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4. Question 12 · part b · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

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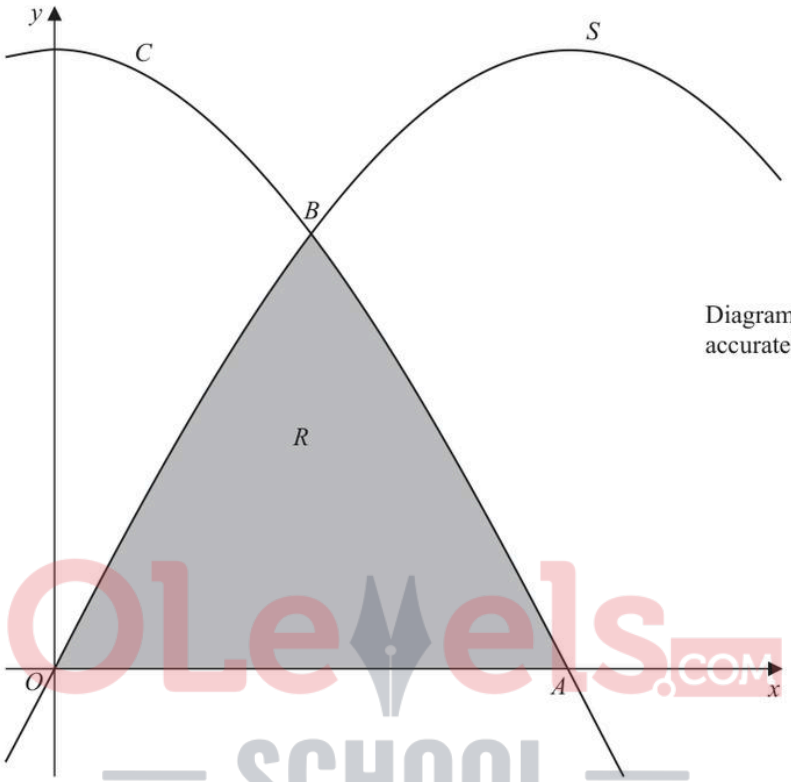


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the x-axis, the curve S with equation $y = 2\sin x$ and the curve C with equation $y = 2\cos x$.
As shown in Figure 2, C crosses the x-axis at the point A.

(a) Write down the x coordinate of A.

(1)

As shown in Figure 2, C and S intersect at the point B.

(b) Find the x coordinate of B.

(2)

(c) Using calculus, find the area of the shaded region R.
Give your answer in the form $a - \sqrt{b}$ where a and b are integers.

(4)

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P 6 2 2 8 3 A 0 3 8 4 0

4. Question 12 · part b · 2 marks · Working and answer

Working and answer

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5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

(4)

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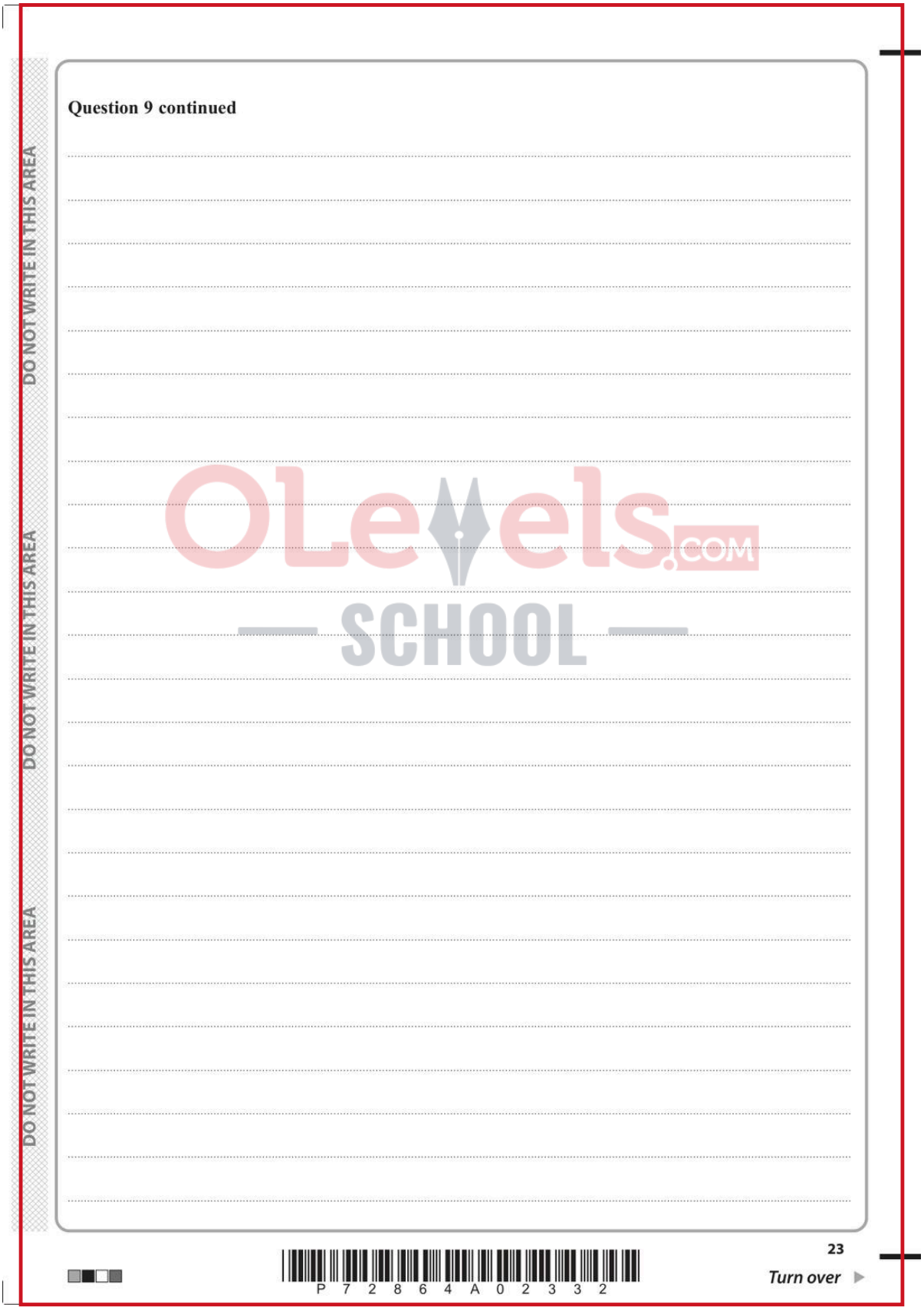
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P 7 2 8 6 4 A 0 2 2 3 2

5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.



5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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5. Question 9 · part c · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

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(Total for Question 9 is 13 marks)

5. Question 9 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 11 · part b · 5 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

11 $f(\theta) = (2\cos\theta - \sin\theta)(2\sin\theta + \cos\theta)$

(a) Show that $f(\theta) = \frac{3}{2}\sin 2\theta + 2\cos 2\theta$

(3)

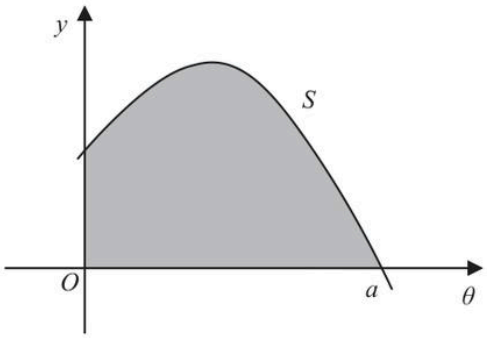


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of the curve S with equation $y = f(\theta) + 2$

Given that S intersects with the θ-axis at the point with coordinates $(a, 0)$

(b) using $\sin^2\theta + \cos^2\theta = 1$, or otherwise, show that $a = \frac{\pi}{2}$

(5)

(c) Using algebraic integration, find the exact area bounded by S, the positive θ-axis and the positive y-axis shown shaded in Figure 4

(3)

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6. Question 11 · part b · 5 marks · Working and answer

Working and answer

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7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.

8 (i) (a) Using a formula given on page 2, show that

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

(2)

(b) Hence, solve the equation

$$\tan A° - \tan 2A° = 0 \qquad \text{for } 0 \leqslant A \leqslant 180$$

(5)

(ii) Using a formula given on page 2, solve, giving your solutions as exact values

$$\cos\left(x - \frac{\pi}{6}\right) = \sin x \qquad \text{for } -\pi \leqslant x \leqslant 2\pi$$

(4)

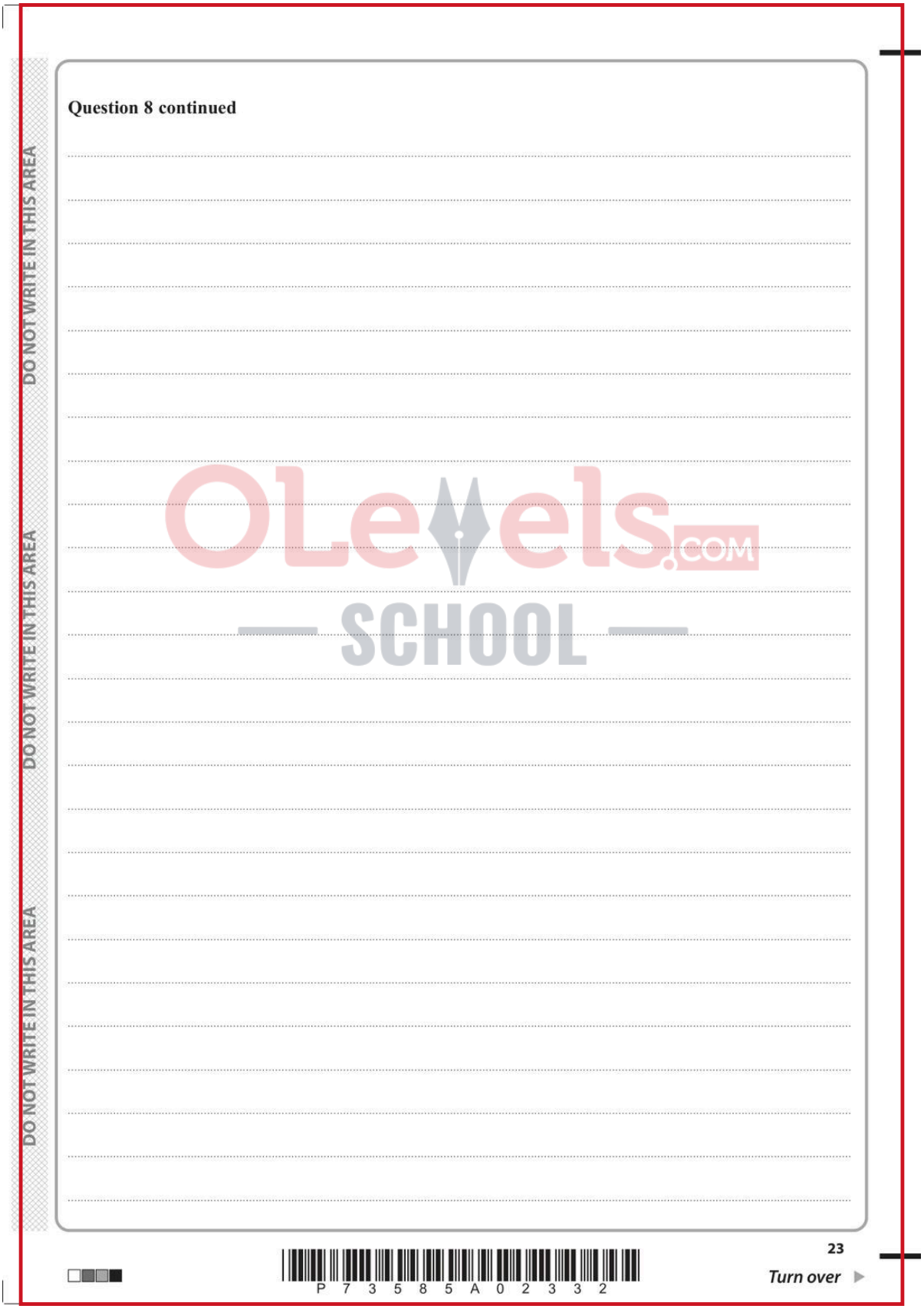
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7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.



Question 8 continued

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Turn over

7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.

Question 8 continued

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7. Question 8 · part b.ii · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b.ii only.

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Question 8 continued

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(Total for Question 8 is 11 marks)

7. Question 8 · part b.ii · 4 marks · Working and answer

Working and answer

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8. Question 7 · part b.ii · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part b.ii only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

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8. Question 7 · part b.ii · 4 marks · Working and answer

Working and answer

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9. Question 11 · part a · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

11 (a) Express the equation

$$3\sin(A - B) = \sin(A + B)$$

in the form $\tan A = k \tan B$, giving the value of the integer k.

(4)

(b) Given that $\theta \neq \frac{(2n+1)\pi}{2}$ where $n \in \mathbb{Z}$,

show that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} = 1 - \tan^2\theta$

(3)

(c) Using the exact values of $\sin x°$, $\cos x°$ and $\tan x°$ for $x = 30, 45, 60$

show that

(i) $\cos 15° = \frac{\sqrt{6} + \sqrt{2}}{4}$

(2)

(ii) $\tan 255° = \frac{3 + \sqrt{3}}{3 - \sqrt{3}}$

(4)

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9. Question 11 · part a · 4 marks · Working and answer

Working and answer

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Practice workbook · Trig equations that factorise

Solving an equation by factorising out a common trig factor (or using a product form) and solving each factor, instead of dividing and losing solutions.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-26

1. Question 8 · part b · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

8 (a) Solve, to the nearest integer, the equation

$$\sin(2x - 75)° = -0.515 \quad \text{for } 0 \leqslant x < 180$$

(3)

(b) Giving your solutions to one decimal place, where appropriate, solve the equation

$$2\tan y° + 5\sin y° = 0 \quad \text{for } 0 \leqslant y \leqslant 180$$

(4)

(c) Explain mathematically why there are no values of θ that satisfy the equation

$$3\cos^2\theta° - 3\sin^2\theta° + \sin\theta° + 12 = 0$$

(4)

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1. Question 8 · part b · 4 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

1 Solve the equation

$$\sqrt{3}\tan A = 2\sin A \quad \text{for} \quad 0^\circ \leqslant A \leqslant 360^\circ$$

(5)

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(Total for Question 1 is 5 marks)

2. Question 1 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part b · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 (a) Solve the equation

$$\cos(3x - 15)° = \frac{\sqrt{3}}{2} \quad \text{for} \quad 0 \leqslant x < 180$$

(4)

(b) Solve, giving your solutions to one decimal place where appropriate,

$$3\tan y° + 4\sin y° = 0 \quad \text{for} \quad -180 \leqslant y < 180$$

(4)

(c) Solve, giving your solutions to one decimal place where appropriate,

$$\cos\theta° = 3\sin^2\theta° - 1 \quad \text{for} \quad -180 \leqslant \theta < 180$$

(4)

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3. Question 7 · part b · 4 marks · Working and answer

Working and answer

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

Question 10 continued

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

Question 10 continued

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 13 marks)

4. Question 10 · part c · 4 marks · Working and answer

Working and answer

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5. Question 10 · part c · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

10 (a) Show that

$$\cos(A + B) + \cos(A - B) = 2\cos A \cos B$$

(2)

(b) Hence show that

$$\cos P + \cos Q = 2\cos\frac{P + Q}{2}\cos\frac{P - Q}{2}$$

(3)

(c) Solve, for $0 \leqslant \theta \leqslant \frac{\pi}{2}$, the equation

$$\cos 5\theta + \cos 7\theta = 0$$

Give each solution in terms of π

(4)

(d) Show that

$$\cos 8x + 2\cos 6x + \cos 4x = 4\cos 6x \cos^2 x$$

(3)

(e) Use calculus to find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx$$

(4)

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5. Question 10 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 8 · part c · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

8 (a) Write down the value of k such that $\sin 2A = k \sin A \cos A$

(1)

$$g(A) = 2 + 3\cos A - \sin A - 3\sin 2A - 2\cos^2 A$$

Given that $g(A)$ can be written in the form $(p\cos A - \sin A)(q - r\sin A)$ where p, q and r are integers,

(b) find the value of p, the value of q and the value of r.

(3)

(c) Hence solve, in radians to 3 significant figures where appropriate, the equation

$$g(2\theta) = 0 \quad \text{for} \quad 0 \leqslant \theta < \pi$$

(6)

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6. Question 8 · part c · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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Turn over ▶

6. Question 8 · part c · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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6. Question 8 · part c · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

6. Question 8 · part c · 6 marks · Working and answer

Working and answer

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7. Question 10 · part c · 7 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 Using formulae from page 2

(a) show that $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$

(2)

(b) Hence show that $\cos 5\theta - \cos 9\theta = 2\sin 7\theta \sin 2\theta$

(1)

(c) Solve the equation

$$\cos 5\theta - \cos 9\theta = \sqrt{3}\sin 7\theta \quad \text{for} \quad 0 < \theta \leqslant \frac{1}{3}\pi$$

Give your solutions in terms of π

(7)

(d) Using calculus and showing your working, evaluate

$$\int_0^{\frac{\pi}{7}} 8\sin 7x \cos 2x \tan 2x \, dx$$

Give your answer to 3 decimal places.

(6)

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7. Question 10 · part c · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Quadratic trig equations using an identity

Using sin^2 + cos^2 = 1, a double-angle formula or a proven identity to turn an equation into a quadratic (or polynomial) in one trig ratio, then solving.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-27

1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

10 (a) Solve the equation

$$\tan x^\circ = -3 \qquad \text{for } 0 \leqslant x < 360$$

Give your solutions to the nearest whole number.

(3)

Given that

$$7\sin^2\theta + \sin\theta\cos\theta = 6$$

(b) show that

$$\tan^2\theta + \tan\theta - 6 = 0$$

(3)

(c) Hence solve the equation

$$7\sin^2 y^\circ + \sin y^\circ \cos y^\circ = 6 \qquad \text{for } 0 \leqslant y < 360$$

Give your solutions to the nearest whole number.

(4)

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1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 10 continued

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1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 10 continued

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1. Question 10 · part c · 4 marks

2021 JUN · Paper 02

Worked-example retry · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 10 marks)

1. Question 10 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 11 · part b · 4 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

11 (a) Using a formula from page 2, show that $\cos 2x = 1 - 2\sin^2 x$

(3)

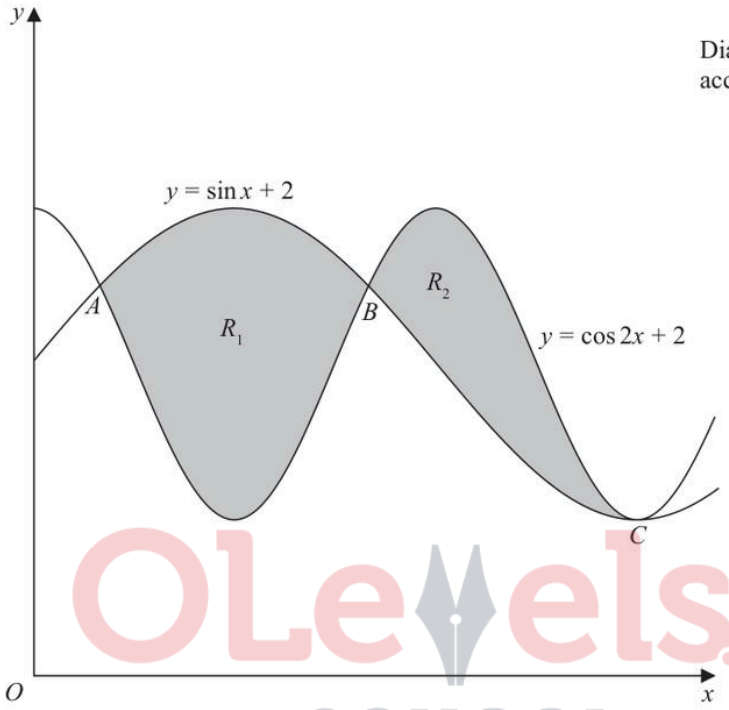


Figure 2

Figure 2 shows a sketch of part of the curves with equations $y = \sin x + 2$ and $y = \cos 2x + 2$

The points A, B and C, shown in Figure 2, are three points that are common to both curves.

(b) Find the coordinates of each of these points.

(4)

R_1 and R_2, shown shaded in Figure 2, are two regions enclosed by the two curves.

(c) Use calculus to find, in its simplest form, the ratio

$$\text{area of } R_1 : \text{area of } R_2$$

(8)

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2. Question 11 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 10 · part b · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \frac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \frac{\cos 2A+1}{2}$

(5)

Diagram **NOT** accurately drawn

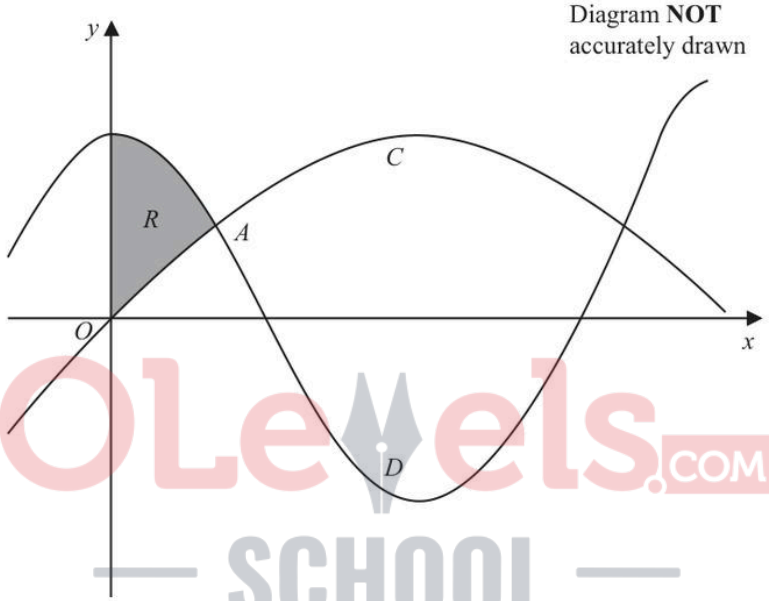


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$
Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through $360°$ about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\frac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

(5)

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P 7 6 5 0 7 A 0 2 6 3 2

3. Question 10 · part b · 5 marks · Working and answer

Working and answer

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4. Question 11 · part b · 7 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

11 (a) Show that $\cos^4\theta - \sin^4\theta = \cos 2\theta$

(4)

(b) Hence, or otherwise, solve the equation

$$8\cos^2\left(2\theta + \frac{\pi}{4}\right) - 3 = 2\cos^4\left(\theta + \frac{\pi}{8}\right) - 2\sin^4\left(\theta + \frac{\pi}{8}\right) \quad \text{for} \quad 0 \leqslant \theta < \pi$$

Give your solutions to 2 decimal places.

(7)

(c) Using calculus, find the exact value of $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \left(\cos^4 2x - \sin^4 2x - 8\sin 4x\right) \mathrm{d}x$

Give your answer in the form $a - b\sqrt{2}$ where a and b are rational numbers.

(4)

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4. Question 11 · part b · 7 marks · Working and answer

Working and answer

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5. Question 11 · part b · 7 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

11 (a) Using a formula on page 2, show that

$$\frac{1-\cos 2A}{1+\cos 2A} = \tan^2 A$$

(3)

(b) Hence, or otherwise, solve in degrees to one decimal place

$$\frac{3-3\cos 4x}{1+\cos 4x} + \frac{5\sin 2x}{\cos 2x} = 2 \quad \text{for } -90^\circ < x < 90^\circ$$

(7)

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5. Question 11 · part b · 7 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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5. Question 11 · part b · 7 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 10 marks)

5. Question 11 · part b · 7 marks · Working and answer

Working and answer

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6. Question 11 · part c · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$

(3)

(ii) $\sin 2A = 2\sin A \cos A$

(1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$

(4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$

(4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$

(4)

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6. Question 11 · part c · 4 marks · Working and answer

Working and answer

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7. Question 8 · part b · 5 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

8 (i) (a) Using a formula given on page 2, show that

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

(2)

(b) Hence, solve the equation

$$\tan A° - \tan 2A° = 0 \qquad \text{for } 0 \leqslant A \leqslant 180$$

(5)

(ii) Using a formula given on page 2, solve, giving your solutions as exact values

$$\cos\left(x - \frac{\pi}{6}\right) = \sin x \qquad \text{for } -\pi \leqslant x \leqslant 2\pi$$

(4)

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7. Question 8 · part b · 5 marks · Working and answer

Working and answer

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8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

7 (a) Solve the equation

$$\cos(3x - 15)° = \frac{\sqrt{3}}{2} \quad \text{for} \quad 0 \leqslant x < 180$$

(4)

(b) Solve, giving your solutions to one decimal place where appropriate,

$$3\tan y° + 4\sin y° = 0 \quad \text{for} \quad -180 \leqslant y < 180$$

(4)

(c) Solve, giving your solutions to one decimal place where appropriate,

$$\cos\theta° = 3\sin^2\theta° - 1 \quad \text{for} \quad -180 \leqslant \theta < 180$$

(4)

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8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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8. Question 7 · part c · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.



8. Question 7 · part c · 4 marks · Working and answer

Working and answer

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9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

10 (a) Solve the equation

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} \qquad \text{for } 0 \leqslant x \leqslant 2\pi$$

Give your solutions in terms of π, where appropriate.

(3)

(b) Solve the equation

$$3\sin\theta + 5\cos\theta = 0 \qquad \text{for } -360° \leqslant \theta \leqslant 360°$$

Give your solutions to the nearest degree.

(3)

(c) Solve the equation

$$1 + \sin 2y = 2\cos^2 2y \qquad \text{for } -180° \leqslant y \leqslant 0°$$

(5)

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9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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9. Question 10 · part c · 5 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 11 marks)

9. Question 10 · part c · 5 marks · Working and answer

Working and answer

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10. Question 10 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

10 (a) Solve the equation

$$\tan x^\circ = -3 \qquad \text{for } 0 \leqslant x < 360$$

Give your solutions to the nearest whole number.

(3)

Given that

$$7\sin^2\theta + \sin\theta\cos\theta = 6$$

(b) show that

$$\tan^2\theta + \tan\theta - 6 = 0$$

(3)

(c) Hence solve the equation

$$7\sin^2 y^\circ + \sin y^\circ \cos y^\circ = 6 \qquad \text{for } 0 \leqslant y < 360$$

Give your solutions to the nearest whole number.

(4)

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10. Question 10 · part b · 3 marks · Working and answer

Working and answer

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11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

8 (a) Solve, to the nearest integer, the equation

$$\sin(2x - 75)° = -0.515 \quad \text{for } 0 \leqslant x < 180$$

(3)

(b) Giving your solutions to one decimal place, where appropriate, solve the equation

$$2\tan y° + 5\sin y° = 0 \quad \text{for } 0 \leqslant y \leqslant 180$$

(4)

(c) Explain mathematically why there are no values of θ that satisfy the equation

$$3\cos^2\theta° - 3\sin^2\theta° + \sin\theta° + 12 = 0$$

(4)

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11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

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Question 8 continued

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11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

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Question 8 continued

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11. Question 8 · part c · 4 marks

2020 JAN · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 11 marks)

11. Question 8 · part c · 4 marks · Working and answer

Working and answer

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